

Electronic structure of Bi₂Te₃ thermoelectric materials

Abstract: We report density functional theory calculations of the density of states (DOS) of Bi₂Te₃. The DOS shows a pronounced peak just below the Fermi level, consistent with its large Seebeck coefficient. The compound exhibits a valence-heavy asymmetry near the Fermi level of approximately 0.3 eV.

Introduction: Bismuth telluride is a well-known thermoelectric material with a high figure of merit at room temperature. Doping with antimony shifts the Fermi level significantly.

Results: The computed $N(E_F)$ is moderate at 0.6 states/eV. A van Hove singularity appears at 0.4 eV below the Fermi level.